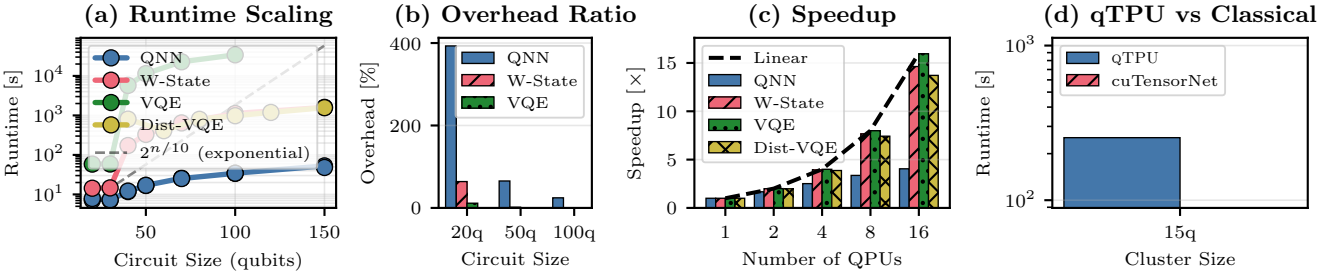
(a) Runtime Scaling
Runtime [s]
$10^4$
$10^3$
$10^2$
$10^1$
QNN
W-State
VQE
Dist-VQE
$2^{n/10}$ (exponential)
50
100
150
Circuit Size (qubits)
(b) Overhead Ratio
Overhead [%]
400
200
0
QNN
W-State
VQE
20q
50q
100q
Circuit Size
(c) Speedup
Speedup [×]
15
10
5
0
Linear
QNN
W-State
VQE
Dist-VQE
1
2
4
8
16
Number of QPUs
(d) qTPU vs Classical
Runtime [s]
$10^3$
$10^2$
qTPU
cuTensorNet
15q
Cluster Size